Week 2 Presentation Content

Data quality, MySQL and KNIME integration

Presentation goal

Demonstrate a validated dataset, working MySQL-KNIME connection, reproducible preparation and a modelling-ready chronological split.

Deck map

Slide	Purpose
1	Weekly status and time review
2	Dataset and decision questions
3	Data quality results
4	MySQL schema and import
5	SQL evidence and KPIs
6	KNIME integration and preparation
7	Initial findings
8	Paper progress and Week 3 plan

Evidence to keep open during Q and A

- python_sql_scripts/results/03_quality_checks_results.xlsx
- sql/ and python_sql_scripts/ for schema, import and quality checks
- knime/ plus screenshots/knime/movielens_like_prediction.png

How to use this file Copy the on-slide text into PowerPoint. Use the visual direction to build the graphic, and rehearse from the speaker cue. Update blank actual-effort fields before presenting.

Slide 1 Week 2 converts raw files into governed analytical data

On-slide copy

- WHAT: validate the data, load MySQL, connect KNIME, prepare features and define the chronological split.
- WHEN: complete quality evidence and modelling tables before Week 3 experiments begin.
- WHY: model metrics are not credible unless the inputs, keys and split logic can be reproduced.
- PLANNED EFFORT: 16-20 team hours. ACTUAL EFFORT: _____ hours. GAP: _____ hours.
- ACTION FROM GAP: _____

Visual direction Use a progress bar from raw files to modelling-ready tables.

Speaker cue Report work completed against the Week 1 baseline. Do not estimate actual hours during the presentation.

Slide 2 The dataset supports four decisions

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- Audience: Which stable user segments can guide curation?
- Engagement: Which user-title pairs are likely to produce ratings of 4.0 or higher?
- Portfolio: Which titles combine high predicted appeal with lower historical popularity?
- Governance: Which results have enough coverage, support and model quality for action?
- Core tables: ratings, movies, tags and a normalized movie_genres bridge.

Visual direction Use a central schema surrounded by four decision questions.

Speaker cue Keep the business questions visible before discussing technical cleaning steps.

Slide 3 Quality checks found no record-level defects

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- Missing, orphan and invalid-value checks returned zero affected ratings and zero affected tags.
- Duplicate user-movie ratings returned no rows; every rating user had at least 20 ratings.
- 10,677 of 10,681 catalogue movies have ratings, leaving four catalogue titles without rating activity.
- No rows were deleted or imputed. Before and after row counts are unchanged.
- Transformations created the genre bridge, chronological split and TRAIN-only aggregate features.

Check	Result	Treatment
Ratings	10,000,054 before and after	No deletion or imputation
Orphan ratings	0	No corrective action
Orphan tags	0	No corrective action
Invalid ratings or timestamps	0	No corrective action
Catalogue coverage	10,677 of 10,681 rated	Retain four unrated titles

Visual direction Use a before-and-after quality table with green zero-defect indicators and one note for four unrated titles.

Speaker cue Separate validation from transformation: the data passed integrity checks, while modelling still required leakage-safe restructuring.

Evidence 03_quality_checks_results.xlsx and the executed SQL outputs.

Slide 4 The MySQL design preserves keys and reusable outputs

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- ratings uses user_id plus movie_id as the natural interaction key and stores rating and timestamp.
- movies stores title and parsed release year; movie_genres resolves the multi-value genre field.
- tags stores free-text annotations linked to users and movies.
- Indexes support joins and time-based filtering; analytical result tables remain separate from raw data.

Visual direction Show a compact relational diagram: movies 1-to-many ratings, users 1-to-many ratings, movies many-to-many genres.

Speaker cue Explain why normalized genres improve filtering and aggregation in both KNIME and Power BI.

Slide 5 SQL makes the evidence traceable

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- SELECT and WHERE validate ranges, timestamps and inclusion rules.
- GROUP BY produces rating distributions, activity counts and genre KPIs.
- JOIN connects ratings to movies and genres while preserving referential integrity.
- Dashboard views expose stable business fields instead of repeating complex joins in Power BI.
- Core KPIs: 10.00M ratings, 69,878 rating users, 10,681 movies, average rating 3.512 and 50.06% positive ratings.

Visual direction Use one short SQL excerpt next to five KPI tiles. Highlight the join keys, not the full query.

Speaker cue Tie every KPI to the query that produced it so the audience can distinguish evidence from interpretation.

Slide 6 KNIME reads from and writes to MySQL

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- Database Connector and Reader nodes pull governed tables from MySQL.
- Preparation nodes enforce types, create the positive-rating target and assemble model fields.
- The workflow creates chronological TRAIN, VALID and TEST partitions rather than a random split.
- Database Writer nodes persist segments, predictions and metrics for dashboard use.
- Credentials remain outside version control; the repository keeps reproducible workflow logic and setup notes.

Visual direction Use a KNIME node-flow screenshot with arrows labeled read, transform, split and writeback.

Speaker cue Demonstrate the live connection briefly, then return to the documented pipeline rather than navigating every node.

Evidence screenshots/knime/movielens_like_prediction.png and repository KNIME documentation.

Slide 7 Initial findings shape the modelling design

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- The target is balanced enough for classification: 50.06% of all ratings are positive.
- User activity is uneven, so the segmentation model must standardize features before K Means.
- The time span allows a realistic out-of-time test rather than evaluating on randomly mixed history.
- Tag volume is small relative to ratings, so tags remain contextual evidence rather than a primary model feature.
- Cold-start coverage must be reported because aggregate features require prior TRAIN history.

Visual direction Use four finding-to-design arrows: balance, scale, time and coverage.

Speaker cue Present these as design implications, not model results; evaluation belongs in Week 3.

Slide 8 Week 3 starts from a frozen modelling contract

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- Paper progress: problem, dataset, architecture and data-quality methodology are ready for the draft.
- Week 3 plan: build TRAIN-only features, tune K Means, fit Logistic Regression and evaluate on TEST.
- Write user_segments, rating_predictions and model_metrics back to MySQL.
- Build an initial five-page Power BI dashboard from versioned result views.
- Carry-forward issue or decision: _____

Effort item	Planned	Actual	Gap and action
Data and SQL	_____ h	_____ h	_____
KNIME preparation	_____ h	_____ h	_____
Documentation	_____ h	_____ h	_____

Visual direction Use a checklist with owners and evidence links. Place the planned-versus-actual time line at the bottom.

Speaker cue Close by naming the exact exit evidence required next week: evaluated models, SQL writeback and dashboard draft.